Assignment 1

Name : Eusha Kayenat

ID : 23201393

Course : CSE471

Section: 09

Ans to the Q&no-1

(a) The four stages of SDLC are-

(I) Planning - Why should the system?

(II) Analysis - Who, what, when, where will be the system?

(III) Design - How will the system work?

(IV) Implementation - How System delivery will be delivered?

(b) (i) Planning phase

(ii) Analysis phase

(iii) Design phase

(iv) Implementation phase

(c) Agile methodology because it has short iterative sprints, continuous stakeholder feedback option which is perfect for requirement changes.

Ans to the Q.no-2

(a) (1) Dev and Ops worked in different isolated environments with different timelines and no synchronization which caused delays.

(2) Testing and deploy used to happen late and manually after the design-build step. It was slow and had errors.

(b) The eight DevOps stages in correct order are-

(1) Plan

(2) Code

③ Build

④ Test

⑤ Release

⑥ Deploy

⑦ Operate

⑧ Monitor

Ans to the Q.no-3

(a) The three categories of Feasibility are-

① Technical feasibility ⇒ Can we built it?

② Economic feasibility ⇒ Should we build it?

③ Organizational feasibility ⇒ If we build it, will they come?

(b) Technical risk :- the new system may be hard to integrate with the existing store systems,

Economic: Development costs may overrun or it will cross the break even point

Organizational: Pharmacists and staff may resist the changed workflow or customers.

Ans to the Q^no-4

Item	Year 1	Year 2	Year 3	Year 4
Total Benefits	150000	200000	250000	300000
Development costs	240000	0	0	0
Operational costs	60000	60000	70000	80000
Total costs	(240000+60000) =300000	(60000+0) =60000	(70000+0) =70000	(80000+0) =80000

Item	Year 1	Year 2	Year 3	Year 4	Total
PV factor	0.909	0.826	0.751	0.683	
PV of Benefits	(150000 X 0.909) = 136350	(200000X0.826) = 165200	(250000X0.751) = 187	(300000X0.683) = 204900	694200
PV of costs	(300000 X 0.909) = 272700	(60000X0.826) = 49560	(70000X0.751) = 52570	(80000X0.683) = 54640	429470

(b) ΣPV Benefits = 694200

ΣPV costs = 429470

NPV = (ΣPV Benefits - ΣPV costs)

= (694200 - 429470)

= 264730

(c) $ROI = \frac{\text{PV Benefits - PV Costs}}{\text{PV costs}}$

$= \frac{(694200 - 429470)}{429470}$

= 0.6164

= 61.64%

(d)

Item	Year 1	Year 2	Year 3	Year 4
Net Benefits	(136350)	115640	135180	150260
Cumulative Net Cash flow	(136350)	(20710)	114470	264730

BEP = Number of years of negative cash flow + $\frac{\text{That years Net Cash flow} - \text{That years cumulative net cash flow}}{\text{That years net cash flow}}$

$= 2 + \frac{(135180 - 114470)}{135180}$

$= 2.15$ years

Ans to the Q.no-5

(a) A System Request for Online Study-Room Reservation system is given below-

① Project Sponsor - The Head Librarian (project champion) is the one who initiates the project and provides resources.

② Business Need - The library needs a more efficient and convenient process. Room reservations causes double bookings or gives students limited access as it is handled at the front desk manually.

③ Business Requirements - The system should let students search availability of rooms by date and time slot, reserve, modify and cancel bookings online.

(4) Business value - improved student satisfaction and a better library reputation

(5) Special Issues - the system must be protected, it should have security and privacy, it must work with the student ID or login system.

(b) (i) Tangible

(ii) Intangible

(iii) Tangible

(iv) Intangible

6) A functional requirement states what the system should do, defines all the services required by the customer that must be provided by the system.

A non-functional requirement states how the software will do a task, it defines system properties and constraints.

(b) i) Functional

2) Non-functional: Performance

3) Non-functional: Security

4) Non-functional: Cultural and Political

Ans to the Q.no-7

(i) Questionnaire - because it reaches a large group easily and quickly

(ii) JAD session - because it brings all stakeholders together to resolve conflicts and agree on requirement.

(iii) Observation - because we can see if people really do as they say

Ans to the Q:no-8

(a) Total processing time = (15+30+20+25+10) = 100 minut

Total elapsed time is 6 days (8-hour days)

∴ Total elapsed time = (6×8×60) minutes

= 2880 minutes

The gap between elapsed time and processed time =

(2880-100) min

= 2780 minutes

(b) If "verify document" and "credit check" run in parallel, then together they'll take 30 minutes to complete the tasks.

New Processing Time = (15+30+25+10) minutes

= 80 minutes

Saved Time = (100 -80) minutes

= 20 minutes

(c) (i) Problem Analysis

(ii) Root cause Analysis